

Optimization Techniques for Machine Learning

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Abstract

You have already studied the classical toolbox of optimization: gradients, Hessians, optimality conditions, gradient descent, and Newton-type methods. In these notes we begin the *machine-learning* part of the course. We first recall what a neural network is and why training it is an optimization problem. We then develop the theory of convexity needed to understand *which* learning problems are “easy” (every local minimizer is global) and which are not. Along the way we prove that a quadratic function is convex exactly when its matrix is positive semidefinite, that local minimizers of convex programs are global, we derive the least-squares objective and the normal equations for linear regression, and we write down the logistic (cross-entropy) loss with its gradient.

1 Setting the Stage: Learning as Optimization

1.1 Supervised learning in one paragraph

We are given a *training set* $\{(x_i, y_i)\}_{i=1}^m$ with inputs $x_i \in \mathbb{R}^n$ (*features*) and targets y_i (a real number in *regression*; a class label such as 0 or 1 in *binary classification*). We choose a parametrized family of prediction functions $x \mapsto f_\theta(x)$, where $\theta \in \mathbb{R}^p$ collects the *parameters* (weights), and a *loss function* $\ell(f_\theta(x), y) \geq 0$ that measures the error of a single prediction. Learning means solving the **empirical risk minimization (ERM)** problem

$$\min_{\theta \in \mathbb{R}^p} J(\theta) = \frac{1}{m} \sum_{i=1}^m \ell(f_\theta(x_i), y_i) \text{ (+ regularizer)}. \quad (1)$$

This is an unconstrained (sometimes constrained) optimization problem in θ , so everything you learned in the first part of the course applies. The new questions are: *what structure does J have?* and *what can we guarantee about the points an algorithm finds?*

1.2 What is a neural network?

Definition 1.1 (Artificial neuron). Let $x \in \mathbb{R}^n$ be an input, $w \in \mathbb{R}^n$ a weight vector, $b \in \mathbb{R}$ a bias and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ an *activation function*. The neuron computes

$$a = \varphi(w^\top x + b).$$

Common activations are the identity $\varphi(z) = z$, the sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$, the hyperbolic tangent, and the ReLU (Rectified Linear Unit) $\varphi(z) = \max\{0, z\}$.

Definition 1.2 (Feed-forward neural network). A feed-forward network with L layers is the composition

$$h^{(0)} = x, \quad h^{(k)} = \varphi_k(W^{(k)}h^{(k-1)} + b^{(k)}), \quad k = 1, \dots, L, \quad f_\theta(x) = h^{(L)},$$

where $W^{(k)}$ are weight matrices, $b^{(k)}$ bias vectors, and $\theta = \{W^{(k)}, b^{(k)}\}_{k=1}^L$ is the full parameter set. The activation φ_k acts entrywise.

Training a network means solving (1) for this f_θ . The gradient $\nabla J(\theta)$ is computed by the chain rule applied layer by layer (*backpropagation*), and then used inside gradient descent or its stochastic variants. We will study these algorithms later in the course. First we must understand the geometry of J .

Two single-neuron networks that we will analyse completely

1. **Linear regression:** one neuron, identity activation, squared-error loss, $f_w(x) = w^\top x$.
2. **Logistic regression:** one neuron, sigmoid activation, cross-entropy loss, $f_w(x) = \sigma(w^\top x)$.

Both are *convex*. A genuine multi-layer network with nonlinear hidden activations is generally *not*, because compositions of convex functions with nonlinear maps need not be convex. Understanding why the single neuron is convex, and what convexity buys us, is the foundation for understanding what can and cannot be guaranteed for deep networks.

2 Mathematical Preliminaries

2.1 Notation

Vectors are columns in $\mathbb{R}^n$; $x^\top y = \langle x, y \rangle$ is the Euclidean inner product and $\|x\| = \sqrt{x^\top x}$. For a differentiable $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla f(x) \in \mathbb{R}^n$ is the gradient; for twice differentiable f , $\nabla^2 f(x) \in \mathbb{R}^{n \times n}$ is the (symmetric) Hessian. $\mathbb{S}^n$ denotes the real symmetric $n \times n$ matrices.

2.2 Positive semidefinite matrices

Definition 2.1. A symmetric matrix $Q \in \mathbb{S}^n$ is

- **positive semidefinite** (PSD, $Q \succeq 0$) if $v^\top Q v \geq 0$ for all $v \in \mathbb{R}^n$;
- **positive definite** (PD, $Q \succ 0$) if $v^\top Q v > 0$ for all $v \neq 0$.

By the spectral theorem, $Q = U \Lambda U^\top$ with U orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Writing $u = U^\top v$ gives $v^\top Q v = \sum_i \lambda_i u_i^2$, hence

$$Q \succeq 0 \iff \lambda_i \geq 0 \forall i, \quad Q \succ 0 \iff \lambda_i > 0 \forall i.$$

Moreover, $v^\top Q v \geq \lambda_{\min}(Q) \|v\|^2$ for every v .

Lemma 2.2 (Gram matrices are PSD). *For any $B \in \mathbb{R}^{k \times n}$, the matrix $B^\top B$ is symmetric PSD, and $B^\top B \succ 0$ iff $\text{rank } B = n$.*

Proof. $v^\top B^\top B v = \|Bv\|^2 \geq 0$. Equality holds iff $Bv = 0$; so $v^\top B^\top B v > 0$ for all $v \neq 0$ iff $\text{null}(B) = \{0\}$, i.e. $\text{rank } B = n$. $\square$

2.3 Convex sets and convex functions

Definition 2.3 (Convex set). $C \subseteq \mathbb{R}^n$ is convex if $x, y \in C$ and $\lambda \in [0, 1]$ imply $\lambda x + (1 - \lambda)y \in C$.

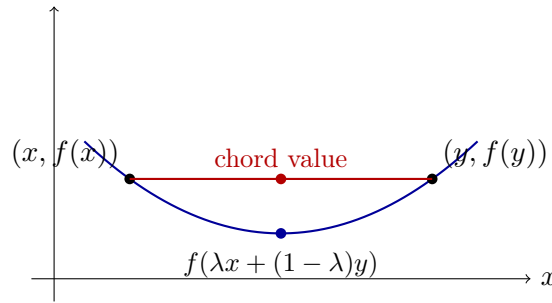
Definition 2.4 (Convex function). Let $C \subseteq \mathbb{R}^n$ be convex. A function $f : C \rightarrow \mathbb{R}$ is

- **convex** if for all $x, y \in C$ and $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y); \quad (2)$$

- **strictly convex** if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$;
- **μ -strongly convex** ($\mu > 0$) if $f - \frac{\mu}{2} \|\cdot\|^2$ is convex.

Geometrically, (2) says that the chord joining two points of the graph lies on or above the graph.



Lemma 2.5 (First-order characterization). *Let C be open and convex and $f : C \rightarrow \mathbb{R}$ be differentiable and convex. Then for all $x, y \in C$,*

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

Consequently, if $\nabla f(x^*) = 0$ then x^* is a global minimizer of f on C .

Proof. For $t \in (0, 1]$, convexity gives $f(x + t(y - x)) \leq (1 - t)f(x) + tf(y)$, i.e.

$$\frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

Letting $t \downarrow 0$ the left side tends to $\nabla f(x)^\top (y - x)$, which proves the inequality. If $\nabla f(x^*) = 0$ it reads $f(y) \geq f(x^*)$ for all y . $\square$

3 Convexity of Quadratic Functions

Quadratic functions are the simplest non-trivial objectives, and they are the building block of second-order (Newton) analysis, of least squares, and of the local behaviour of every smooth loss. We therefore characterize precisely when they are convex.

Theorem (Convexity of quadratic forms)

Let $Q \in \mathbb{S}^n$ and $f(x) = \frac{1}{2}x^\top Qx$. Then

$$f \text{ is convex on } \mathbb{R}^n \iff Q \succeq 0.$$

Moreover, f is strictly convex $\iff Q \succ 0$, in which case f is $\lambda_{\min}(Q)$ -strongly convex.

Remark 3.1 (Why symmetric Q is no loss of generality). If Q is not symmetric, then $x^\top Qx = x^\top (\frac{Q+Q^\top}{2})x$, since $x^\top Qx$ is a scalar and equals its own transpose $x^\top Q^\top x$. So f only “sees” the symmetric part $\frac{1}{2}(Q + Q^\top)$, and we may always assume $Q = Q^\top$. Adding a linear term $c^\top x$ or a constant does not change convexity (see Corollary 3.4 below).

Lemma 3.2 (Key identity). For $f(x) = \frac{1}{2}x^\top Qx$ with $Q = Q^\top$, all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$,

$$\lambda f(x) + (1 - \lambda)f(y) - f(\lambda x + (1 - \lambda)y) = \frac{\lambda(1 - \lambda)}{2} (x - y)^\top Q (x - y).$$

Proof. Set $z = \lambda x + (1 - \lambda)y$. Expanding with symmetry of Q ,

$$z^\top Qz = \lambda^2 x^\top Qx + 2\lambda(1 - \lambda)x^\top Qy + (1 - \lambda)^2 y^\top Qy.$$

Hence

$$\begin{aligned} \lambda x^\top Qx + (1 - \lambda)y^\top Qy - z^\top Qz \\ &= \lambda(1 - \lambda)x^\top Qx + \lambda(1 - \lambda)y^\top Qy - 2\lambda(1 - \lambda)x^\top Qy \\ &= \lambda(1 - \lambda) (x - y)^\top Q (x - y), \end{aligned}$$

where we used $\lambda - \lambda^2 = (1 - \lambda) - (1 - \lambda)^2 = \lambda(1 - \lambda)$. Multiplying by $\frac{1}{2}$ gives the claim. $\square$

Proof of the Theorem. ($\Leftarrow$) Suppose $Q \succeq 0$. For $\lambda \in [0, 1]$ the factor $\lambda(1 - \lambda) \geq 0$ and $(x - y)^\top Q (x - y) \geq 0$, so by Lemma 3.2 the left-hand side is ≥ 0 , which is exactly the convexity inequality (2).

($\Rightarrow$) Suppose f is convex. Let $v \in \mathbb{R}^n$ be arbitrary. Choose $x = v$, $y = 0$ and $\lambda = \frac{1}{2}$ in Lemma 3.2. Convexity makes the left side nonnegative, hence

$$0 \leq \frac{1}{2} \cdot \frac{1}{4} v^\top Qv \implies v^\top Qv \geq 0.$$

As v was arbitrary, $Q \succeq 0$.

Strict convexity. If $Q \succ 0$, then for $x \neq y$ and $\lambda \in (0, 1)$ the right side of Lemma 3.2 is strictly positive, so f is strictly convex. Conversely, if f is strictly convex, the same choice $x = v \neq 0$, $y = 0$, $\lambda = \frac{1}{2}$ gives a strictly positive left side, hence $v^\top Qv > 0$. For strong convexity, note $f(x) - \frac{\mu}{2} \|x\|^2 = \frac{1}{2}x^\top (Q - \mu I)x$, which is convex iff $Q - \mu I \succeq 0$ by the first part, i.e. iff $\mu \leq \lambda_{\min}(Q)$. $\square$

Remark 3.3 (Second-order viewpoint). Since $\nabla f(x) = Qx$ and $\nabla^2 f(x) = Q$ for all x , the theorem is the special case of the general fact: *a twice-differentiable function on a convex open set is convex iff $\nabla^2 f(x) \succeq 0$ for every x .* For a quadratic the Hessian is constant, which is why a single matrix condition decides global convexity. For non-quadratic f the Hessian varies with x and one needs $\nabla^2 f(x) \succeq 0$ everywhere.

Corollary 3.4 (Affine terms do not matter). $g(x) = \frac{1}{2}x^\top Qx + c^\top x + d$ is convex iff $Q \succeq 0$. Indeed $g = f + \ell$ where $\ell(x) = c^\top x + d$ is affine, and the identity of Lemma 3.2 holds with equality for affine ℓ , so it contributes 0 to the left side.

Example 3.5. Let $Q = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Eigenvalues are 1 and 3, both positive, so $f(x) = \frac{1}{2}x^\top Qx = x_1^2 + x_1x_2 + x_2^2$ is strictly convex. For $Q = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ the eigenvalues are 3 and -1 ; the direction $v = (1, -1)^\top$ gives $v^\top Qv = -2 < 0$, so f is not convex (it is a saddle).

4 Convex Programs: Every Local Minimizer Is Global

Definition 4.1 (Convex program). A *convex program* is a problem

$$\min_{x \in C} f(x),$$

where $C \subseteq \mathbb{R}^n$ is a convex set (e.g. $C = \mathbb{R}^n$, or a set defined by linear equalities and convex inequalities) and $f : C \rightarrow \mathbb{R}$ is a convex function.

Definition 4.2 (Local and global minimizers). A point $x^* \in C$ is a *local minimizer* if there is $\varepsilon > 0$ such that $f(x^*) \leq f(x)$ for all $x \in C$ with $\|x - x^*\| < \varepsilon$. It is a *global minimizer* if $f(x^*) \leq f(x)$ for all $x \in C$.

Theorem (Local

Let C be convex and $f : C \rightarrow \mathbb{R}$ convex. Then every local minimizer of f over C is a global minimizer. Moreover, the set of global minimizers is convex, and if f is strictly convex the minimizer (if it exists) is unique.

Proof. Let x^* be a local minimizer with radius $\varepsilon > 0$, and suppose, for contradiction, that it is not global: there exists $y \in C$ with $f(y) < f(x^*)$. For $\lambda \in (0, 1]$ put

$$z_\lambda = \lambda y + (1 - \lambda)x^*.$$

Then $z_\lambda \in C$ because C is convex, and by convexity of f ,

$$f(z_\lambda) \leq \lambda f(y) + (1 - \lambda)f(x^*) < \lambda f(x^*) + (1 - \lambda)f(x^*) = f(x^*).$$

Now $\|z_\lambda - x^*\| = \lambda \|y - x^*\|$, which is less than ε once $\lambda < \varepsilon / \|y - x^*\|$. Choosing such a λ produces a point $z_\lambda \in C$ arbitrarily close to x^* with $f(z_\lambda) < f(x^*)$, contradicting local minimality. Hence x^* is global.

Convexity of the solution set. Let $f^* = \min f$ and x_1, x_2 be global minimizers. Then $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f^* + (1 - \lambda)f^* = f^*$, so every convex combination is again a minimizer.

Uniqueness under strict convexity. If $x_1 \neq x_2$ were both global minimizers, strict convexity would give $f(\frac{x_1 + x_2}{2}) < f^*$, which is impossible. $\square$

Why this matters for optimization algorithms. Gradient descent, Newton's method and their relatives are *local* methods: they can only certify that the current point is stationary ($\nabla f = 0$) or a local minimizer. For convex differentiable f on $\mathbb{R}^n$, Lemma 2.5 says that

$$\nabla f(x^*) = 0 \iff x^* \text{ is a global minimizer.}$$

Thus, for convex problems, “the algorithm has converged to a stationary point” means “the problem is solved”. For the non-convex objectives of deep networks this fails: there are spurious local minimizers and saddle points, and the quality of the solution depends on initialization, step sizes and other algorithmic details.

5 Linear Regression as a Least-Squares Problem

5.1 The model and the objective

Consider the single linear neuron $f_w(x) = w^\top x$ (a bias can be absorbed by appending a constant 1 to every feature vector). Stack the m training inputs as the rows of the **design matrix**

$$A = \begin{pmatrix} x_1^\top \\ \vdots \\ x_m^\top \end{pmatrix} \in \mathbb{R}^{m \times n}, \quad b = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \in \mathbb{R}^m.$$

The vector of predictions is Aw and the residual is $r(w) = Aw - b$. Using the squared-error loss $\ell(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2$ and summing over the data, the **least-squares objective** is

$$J(w) = \frac{1}{2} \|Aw - b\|^2 = \frac{1}{2} \sum_{i=1}^m (x_i^\top w - y_i)^2. \quad (3)$$

(The factor $\frac{1}{2}$ only simplifies the derivative; an additional $\frac{1}{m}$ would rescale J without changing its minimizers.)

5.2 Derivation of the gradient and Hessian

Expand the squared norm:

$$\begin{aligned} J(w) &= \frac{1}{2} (Aw - b)^\top (Aw - b) \\ &= \frac{1}{2} w^\top A^\top Aw - b^\top Aw + \frac{1}{2} b^\top b. \end{aligned} \quad (4)$$

This is exactly of the quadratic form $\frac{1}{2}w^\top Qw + c^\top w + d$ with

$$Q = A^\top A, \quad c = -A^\top b, \quad d = \frac{1}{2} \|b\|^2.$$

For a quadratic with symmetric Q one has $\nabla(\frac{1}{2}w^\top Qw) = Qw$ and $\nabla(c^\top w) = c$. To make this rigorous, perturb w by h :

$$J(w + h) = J(w) + \underbrace{(A^\top Aw - A^\top b)^\top h}_{\text{linear in } h} + \frac{1}{2} h^\top A^\top A h,$$

and the last term is $\frac{1}{2} \|Ah\|^2 = O(\|h\|^2)$. By the definition of the derivative,

$$\boxed{\nabla J(w) = A^\top (Aw - b) = A^\top Aw - A^\top b}, \quad \boxed{\nabla^2 J(w) = A^\top A}. \quad (5)$$

Equivalently, in coordinates, $\nabla J(w) = \sum_{i=1}^m (x_i^\top w - y_i) x_i$: each data point pushes w in the direction of its feature vector, weighted by its residual.

5.3 The normal equations

Because J is convex (proved next) and differentiable, Lemma 2.5 shows that w^* is a global minimizer iff $\nabla J(w^*) = 0$, i.e.

$$\boxed{A^\top A w = A^\top b} \quad (\text{normal equations}). \quad (6)$$

The name comes from the equivalent form $A^\top(b - Aw^*) = 0$: the residual is *normal* (orthogonal) to every column of A , i.e. Aw^* is the orthogonal projection of b onto $\text{range}(A)$.

Proposition 5.1 (Solvability and uniqueness). (a) *The normal equations always have at least one solution, so J always attains its minimum.*
 (b) *The solution is unique iff $\text{rank } A = n$ (the columns of A are linearly independent); then*

$$w^* = (A^\top A)^{-1} A^\top b.$$

(c) *In general the solution set is $w^\dagger + \text{null}(A)$, where $w^\dagger = A^+ b$ (Moore–Penrose pseudoinverse) is the one of minimum Euclidean norm.*

Proof. (a) We know $\text{range}(A^\top A) = \text{range}(A^\top)$: indeed $\text{null}(A^\top A) = \text{null}(A)$, because $A^\top A v = 0 \Rightarrow \|A v\|^2 = v^\top A^\top A v = 0 \Rightarrow A v = 0$; taking orthogonal complements gives $\text{range}(A^\top A) = \text{null}(A^\top A)^\perp = \text{null}(A)^\perp = \text{range}(A^\top)$. Since $A^\top b \in \text{range}(A^\top)$, the system $A^\top A w = A^\top b$ is consistent.

(b) By Lemma 2.2, $A^\top A \succ 0$ iff $\text{rank } A = n$, in which case it is invertible. If $\text{rank } A < n$, then $\text{null}(A^\top A) = \text{null}(A) \neq \{0\}$ and any solution can be shifted by a null-space vector, so it is not unique.

(c) If w_1 solves (6), so does $w_1 + v$ for any $v \in \text{null}(A) = \text{null}(A^\top A)$, and these are all solutions. The minimum-norm element is the one orthogonal to $\text{null}(A)$, which is $A^+ b$. $\square$

Example 5.2. Fit a line $y \approx w_0 + w_1 t$ to the data $(t, y) = (0, 1), (1, 2), (2, 2)$. Then

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad A^\top A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad A^\top b = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Here $\det(A^\top A) = 6 \neq 0$, so $w^* = \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}$. Check: the residual $A w^* - b = (\frac{1}{6}, -\frac{1}{3}, \frac{1}{6})^\top$ satisfies $A^\top(A w^* - b) = (0, 0)^\top$.

5.4 Justification of convexity

Theorem 5.3 (Convexity of the least-squares objective). *For every $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, the function $J(w) = \frac{1}{2} \|A w - b\|^2$ is convex on $\mathbb{R}^n$. It is strictly convex (indeed $\sigma_{\min}(A)^2$ -strongly convex) iff $\text{rank } A = n$.*

Proof. *Route 1 (via the quadratic-form theorem).* By (4), J is a quadratic with matrix $Q = A^\top A$ plus an affine term. By Lemma 2.2, $A^\top A \succeq 0$, since $v^\top A^\top A v = \|A v\|^2 \geq 0$. By Section 3 (and Corollary 3.4), J is convex. The strict/strong statements follow from $A^\top A \succ 0 \iff \text{rank } A = n$ and $\lambda_{\min}(A^\top A) = \sigma_{\min}(A)^2$.

Route 2 (composition). The map $u \mapsto \frac{1}{2} \|u\|^2$ is convex (it is the quadratic with $Q = I \succ 0$), and $w \mapsto A w - b$ is affine. A convex function composed with an affine map is convex: $g(\lambda w_1 + (1 - \lambda) w_2) = \phi(\lambda(A w_1 - b) + (1 - \lambda)(A w_2 - b)) \leq \lambda g(w_1) + (1 - \lambda) g(w_2)$. $\square$

Consequences for training linear regression

- Every stationary point is a global minimizer, so the normal equations give *all* optimal weights.
- Gradient descent $w_{k+1} = w_k - \eta A^\top (A w_k - b)$ with $0 < \eta \leq 1/\sigma_{\max}(A)^2$ converges to a minimizer. Here $L = \lambda_{\max}(A^\top A) = \sigma_{\max}(A)^2$ is the Lipschitz constant of ∇J , since $\nabla^2 J = A^\top A$ is constant.

- If $\text{rank } A = n$, convergence is linear with rate governed by the condition number $\kappa = \sigma_{\max}^2 / \sigma_{\min}^2$ of $A^\top A$. Badly scaled features (large κ) slow gradient descent, which motivates feature normalization and preconditioning.

6 Logistic Regression: the Loss for a Binary Target

6.1 Why not least squares for classification?

For a binary label $y \in \{0, 1\}$ we want the model to output a *probability*. A linear output $w^\top x \in \mathbb{R}$ is unbounded, so we squash it with the **sigmoid**

$$\sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1), \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)), \quad 1 - \sigma(z) = \sigma(-z).$$

(Derivation: $\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \sigma(z) \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z)(1 - \sigma(z))$.) The model is the single sigmoid neuron

$$p_w(x) = \Pr(y = 1 \mid x) = \sigma(w^\top x), \quad \Pr(y = 0 \mid x) = 1 - \sigma(w^\top x).$$

Composing the sigmoid with squared error gives a non-convex objective in w ; instead we use the principle of maximum likelihood.

6.2 Derivation of the loss

For one example, the Bernoulli probability of the observed label is $p^y(1-p)^{1-y}$ with $p = \sigma(w^\top x)$. Assuming independent examples, the likelihood is $\prod_{i=1}^m p_i^{y_i}(1-p_i)^{1-y_i}$. Maximizing it is equivalent to minimizing the average negative log-likelihood, called the **binary cross-entropy (logistic) loss**:

$$J(w) = -\frac{1}{m} \sum_{i=1}^m \left[y_i \log \sigma(x_i^\top w) + (1 - y_i) \log(1 - \sigma(x_i^\top w)) \right] \quad (7)$$

with $y_i \in \{0, 1\}$. Writing $z_i = x_i^\top w$, a simplification using $-\log \sigma(z) = \log(1 + e^{-z}) = \log(1 + e^z) - z$ and $-\log(1 - \sigma(z)) = \log(1 + e^z)$ yields the numerically convenient “softplus” form

$$J(w) = \frac{1}{m} \sum_{i=1}^m \left[\log(1 + e^{z_i}) - y_i z_i \right], \quad z_i = x_i^\top w. \quad (8)$$

(If labels are encoded as $\tilde{y}_i \in \{-1, +1\}$, the same loss reads $\frac{1}{m} \sum_i \log(1 + e^{-\tilde{y}_i z_i})$.) In code one evaluates $\log(1 + e^z)$ with the log-sum-exp trick to avoid overflow.

6.3 The gradient

Differentiate (8) term by term. With $\ell(z, y) = \log(1 + e^z) - yz$,

$$\frac{\partial \ell}{\partial z} = \frac{e^z}{1 + e^z} - y = \sigma(z) - y, \quad \text{and} \quad \frac{\partial z_i}{\partial w} = x_i.$$

By the chain rule,

$$\nabla J(w) = \frac{1}{m} \sum_{i=1}^m (\sigma(x_i^\top w) - y_i) x_i = \frac{1}{m} A^\top (\sigma(Aw) - y) \quad (9)$$

where A is again the design matrix and σ is applied entrywise. Compare with (5): the structure “ $A^\top \times (\text{prediction} - \text{target})$ ” is identical; only the prediction Aw is replaced by $\sigma(Aw)$. This is not a coincidence: both are instances of *generalized linear models* with the “canonical” loss, and the same expression appears at the output layer of a neural network trained by backpropagation (the output error $\delta = \hat{y} - y$).

6.4 Hessian and convexity

Differentiating (9) once more and using $\sigma' = \sigma(1 - \sigma)$,

$$\nabla^2 J(w) = \frac{1}{m} \sum_{i=1}^m \sigma(z_i)(1 - \sigma(z_i)) x_i x_i^\top = \frac{1}{m} A^\top S A, \quad S = \text{diag}(s_i), \quad s_i = \sigma(z_i)(1 - \sigma(z_i)) \in (0, \frac{1}{4}].$$

Proposition 6.1. J in (7) is convex on $\mathbb{R}^n$.

Proof. For any $v \in \mathbb{R}^n$,

$$v^\top \nabla^2 J(w) v = \frac{1}{m} \sum_{i=1}^m s_i (x_i^\top v)^2 \geq 0,$$

since each $s_i > 0$. Hence $\nabla^2 J(w) \succeq 0$ for all w , and J is convex by the second-order criterion (Remark after the quadratic theorem). $\square$

Also $\nabla^2 J \preceq \frac{1}{4m} A^\top A$, so ∇J is Lipschitz with constant $L = \frac{1}{4m} \sigma_{\max}(A)^2$, and gradient descent with $\eta \leq 1/L$ is well behaved.

Remark 6.2 (Caveats you should know). • J is convex but *not* strongly convex in general: $s_i \rightarrow 0$ when $|z_i|$ is large, so the Hessian can become nearly singular far from the origin.

- If the training data are *linearly separable*, then J can be driven to 0 only as $\|w\| \rightarrow \infty$; no minimizer exists. Adding an ℓ_2 regularizer $\frac{\lambda}{2} \|w\|^2$ makes the objective λ -strongly convex and guarantees a unique minimizer.
- Unlike linear regression, $\nabla J(w) = 0$ is a nonlinear system with no closed-form solution; we solve it iteratively (gradient descent, Newton’s method, i.e. *iteratively reweighted least squares*). Convexity ensures that any convergent iteration reaches the global optimum.

7 Summary

	Quadratic $\frac{1}{2}x^\top Qx$	Linear regression	Logistic regression
Objective	$\frac{1}{2}x^\top Qx$	$\frac{1}{2} \ Aw - b\ ^2$	$\frac{1}{m} \sum [\log(1 + e^{z_i}) - y_i z_i]$
Gradient	Qx	$A^\top (Aw - b)$	$\frac{1}{m} A^\top (\sigma(Aw) - y)$
Hessian	Q	$A^\top A$	$\frac{1}{m} A^\top S A$
Convex iff / why	$Q \succeq 0$	always ($A^\top A \succeq 0$)	always ($S \succ 0$)
Optimum	$Qx = 0$	normal equations	iterative solver

Take-home messages

1. A quadratic $\frac{1}{2}x^\top Qx$ is convex iff $Q \succeq 0$; the proof rests on the identity $\lambda f(x) + (1 - \lambda)f(y) - f(z) = \frac{\lambda(1-\lambda)}{2}(x - y)^\top Q(x - y)$.
2. In a convex program, every local minimizer is global: if a better point existed, points on the segment toward it would be better arbitrarily close to x^* .
3. Linear regression is the convex quadratic $\frac{1}{2}\|Aw - b\|^2$ with $\nabla J = A^\top(Aw - b)$, $\nabla^2 J = A^\top A \succeq 0$, and normal equations $A^\top Aw = A^\top b$, always solvable.
4. Logistic regression uses the cross-entropy loss, with gradient $\frac{1}{m}A^\top(\sigma(Aw) - y)$ and a PSD Hessian, so it is convex too.
5. Deep networks lose convexity through nonlinear hidden layers. This is exactly why the second half of this course studies stochastic gradient methods, momentum, adaptive step sizes and initialization: tools for problems where the local-equals-global guarantee no longer holds.